FOR FORMALLY UNRAMIFIED COMPOSITES OF SPACES WITH LCI MIDDLE STEP THE CONORMAL TRANSITIVITY SEQUENCE IS SHORT EXACT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 11457-11470 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 11472-11487, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side; the schemes twin (same label in more-morphisms) uncounted per ruling.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. In-source lemmas transitivity-conormal, universal-thickening-localize, definition-lci; More on Morphisms, lemma transitivity-conormal-lci.

Source reference. Original lemma 06AE: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 591–603. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04FA: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3309–3325. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06BA: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11487–11487. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $Z \to Y \to X$ be formally unramified morphisms of algebraic spaces over S. Assume that $Z \to Y$ is a local complete intersection morphism. The exact sequence*

$$0 \to i^*\mathcal{C}_{Y/X} \to \mathcal{C}_{Z/X} \to \mathcal{C}_{Z/Y} \to 0$$

of Lemma 06AE is short exact.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Choose a scheme U and a surjective étale morphism $U \to X$. Choose a scheme V and a surjective étale morphism $V \to U \times_X Y$. Choose a scheme W and a surjective étale morphism $W \to V \times_Y Z$. By Lemma 04FA the morphisms $W \to V$ and $V \to U$ are formally unramified. Moreover the sequence $i^*\mathcal{C}_{Y/X} \to \mathcal{C}_{Z/X} \to \mathcal{C}_{Z/Y} \to 0$ restricts to the corresponding sequence $i^*\mathcal{C}_{V/U} \to \mathcal{C}_{W/U} \to \mathcal{C}_{W/V} \to 0$ for $W \to V \to U$. Hence the result follows from the result for schemes (More on Morphisms, Lemma 06BA) as by definition the morphism $W \to V$ is a local complete intersection morphism. $\square$